2ND SEMESTER LAB REPORT

TITLE : ANALYSIS OF SERIES AND PARALLEL CIRCUITS



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BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

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Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
FACULTY OF ENGINEERING AIR UNIVERSITY,ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 4

Lab Title: VERIFCATION OF OHM'S LAW

Objective: 1. **ANALYSIS OF SERIES AND parallel combinations of resistors**

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 10/March/2025 Signature:

LABORATORY EXPERIMENT NO. 4

ANALYSIS OF SERIES AND PARALLEL C IRCUITS

OBJECTIVE:

1. To verify total resistance RT in a circuit in which the resistors R1, R2, R3 etc are connected in series and parallel.

2. To develop a formula based on experimental results that gives the resistance RT of resistors connected in series and parallel.

3. To verify experimentally the branch resistances and the total Resistance of series and parallel circuits.

Ohm's Law:

Ohm's Law is the basis of many electrical circuit calculations and is one of the most important theories you will learn: V=IR.

Measuring Current in a Series Circuit Using a Digital Multimeter (DMM)

To measure the current in a series circuit with a **digital multimeter (DMM)**, follow these steps:

1. **Turn Off Power** – Ensure the circuit is powered off before making connections.
2. **Set the DMM** – Turn the dial to the **current (A or mA)** setting.
3. **Break the Circuit** – Since current flows in a single path in series, disconnect a wire at any point in the circuit.

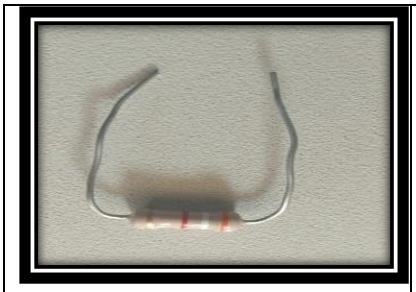
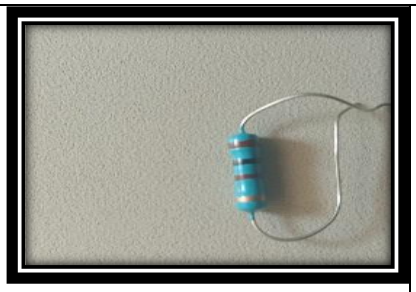
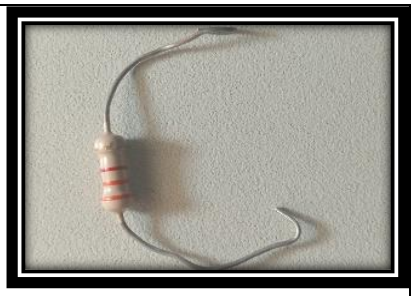
========(Task 1)=========

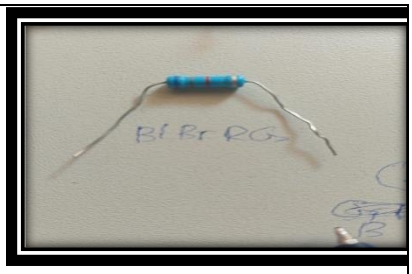
Lab Task:

1. Solve for the following series circuits using ohm-meter:

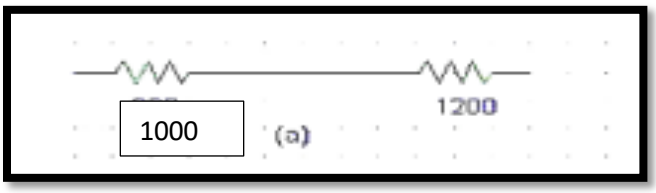
a) Given the nominal values and tolerances in the table given below,determine and record the corresponding color code bands.

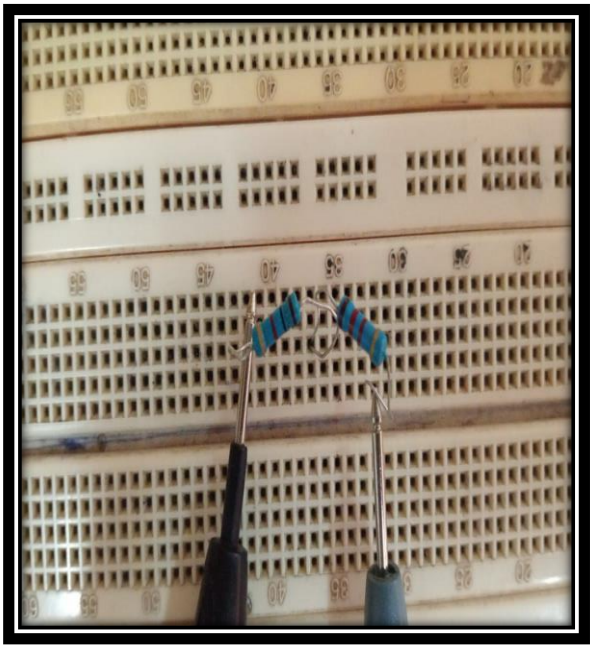
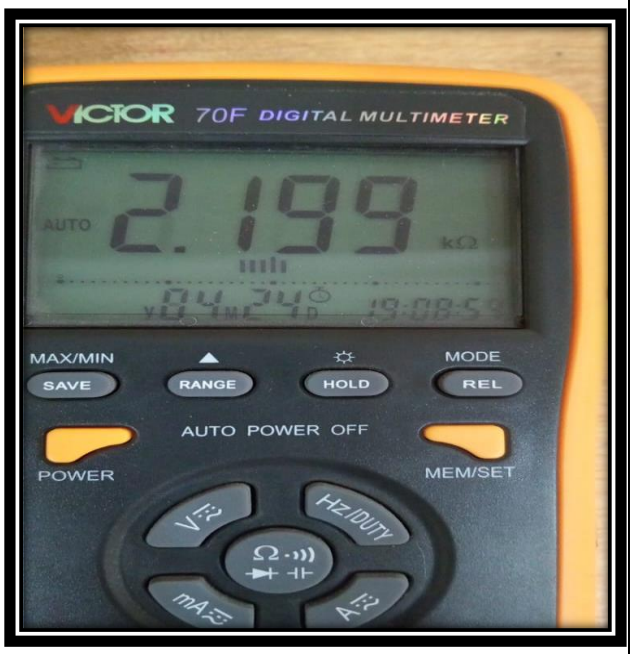
Resistors (Ω)	R1 (Ω)	R2 (Ω)	R3 (Ω)	R4 (Ω)	R5 (Ω)	R6 (Ω)
Band colors	Red-Red Red-Gold	Orange-orange-red-gold	Yellow-violet-red	Yellow-green-blue	Brown-red-red-gold	Brown-black-red-gold
Rated value Ω	2200	3300	470	4700	1200	1000
Measured values Ω	2169	3296	458	4560	1213	0.991

		
2200 Ω	470 Ω	3300 Ω

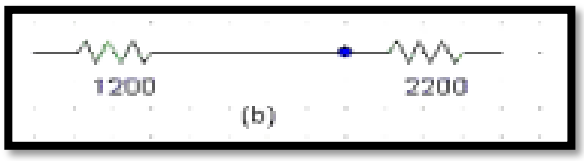
		
4700 Ω	1200 Ω	1000 Ω

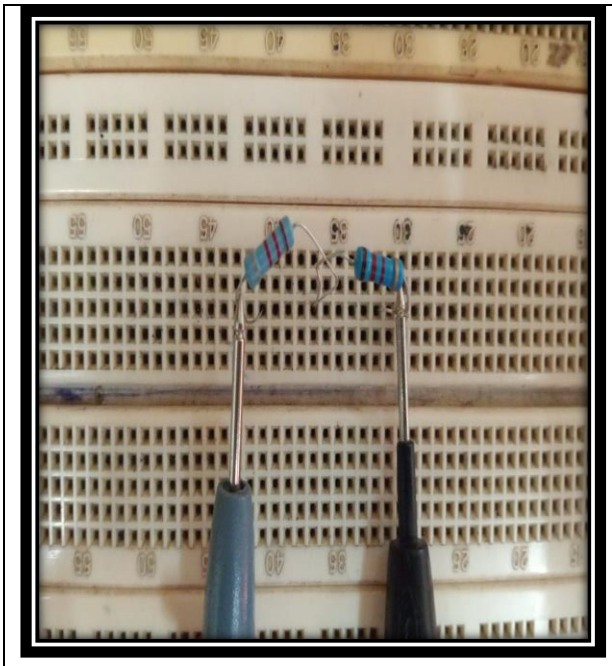
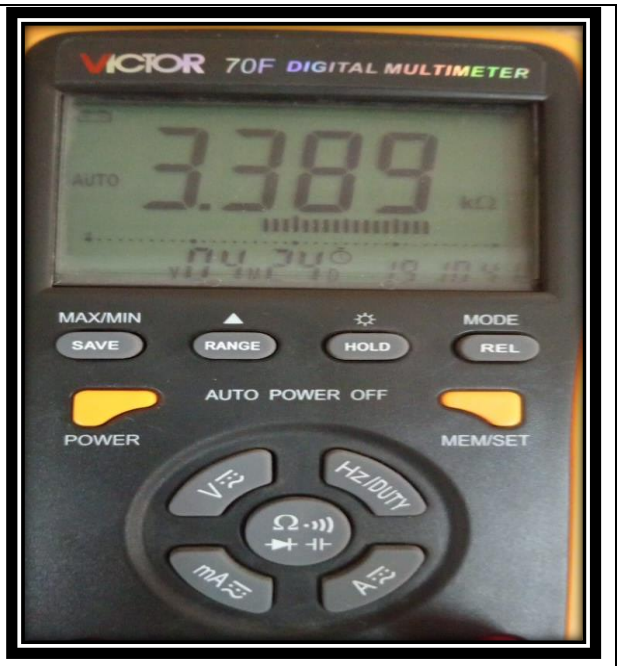
> >>>>>>(a)<<<<<<<



	330
	
This is a mother-board a resitor of 1000 and 1200 ohm	**Multimeter: Show equivalence resitence for resistors connected in series**
Rated value: 2.2k Ω	**Measured Value:**2.199k Ω

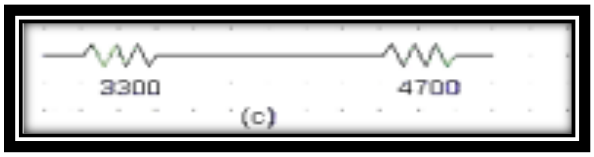
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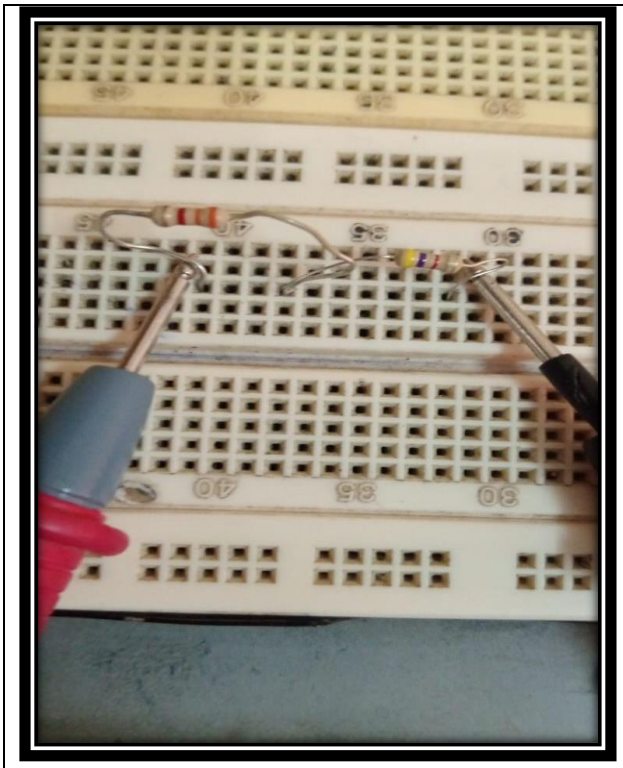
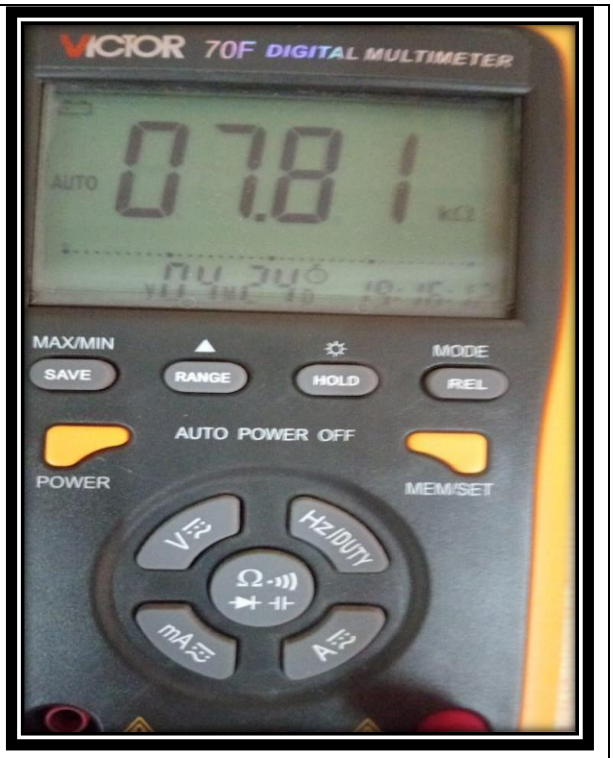


	
Motherboard **R1+R2=1200+2200=3300**	**Mulimeter: Show equivalence resitence for resistors connected in series**
Rated value: 3.3k Ω	**Measured Value:**3.389k Ω

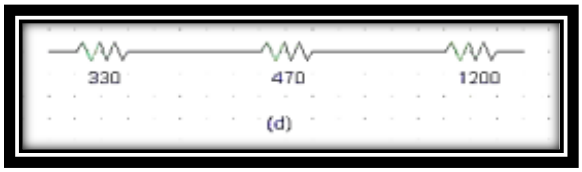
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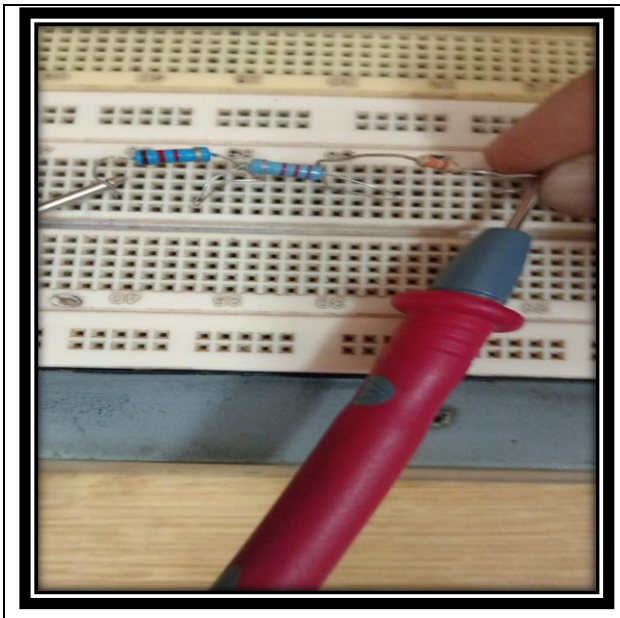
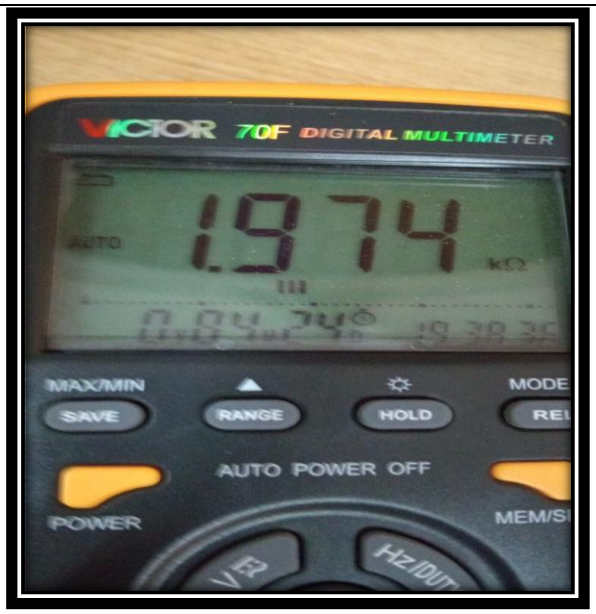
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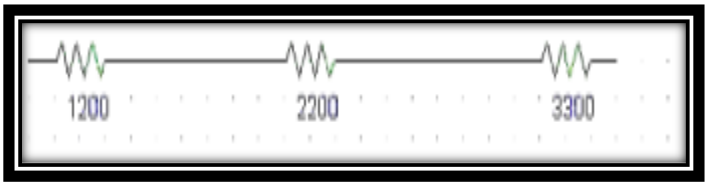
	
Motherboard **R1+R2=3300+4700=8000**	**Mulimeter: Show equivalence resitence for resistors connected in series**
Rated value: 8k Ω	**Measured Value:7.81k Ω**

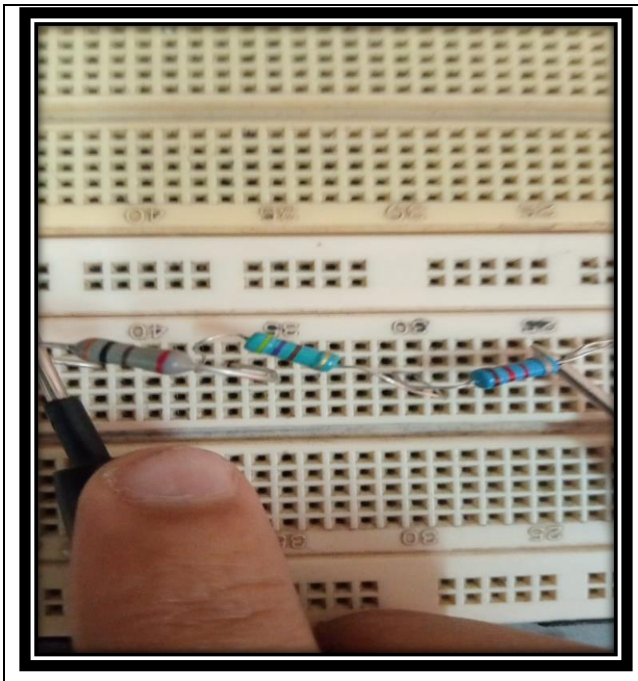
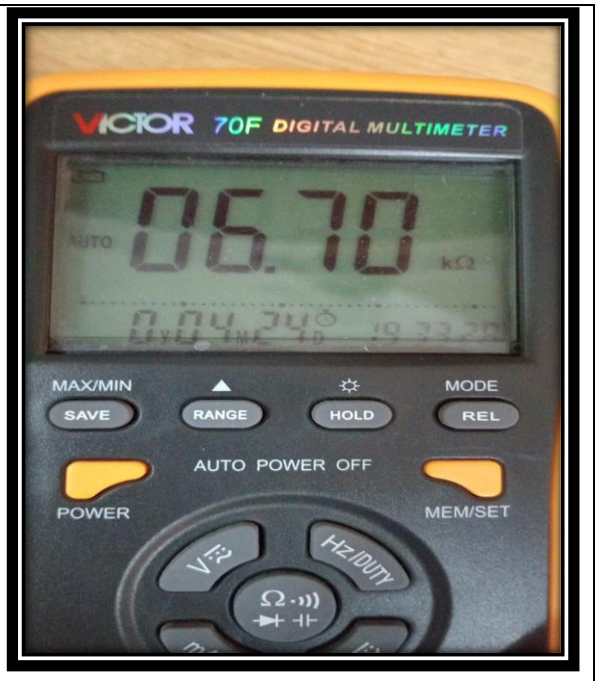
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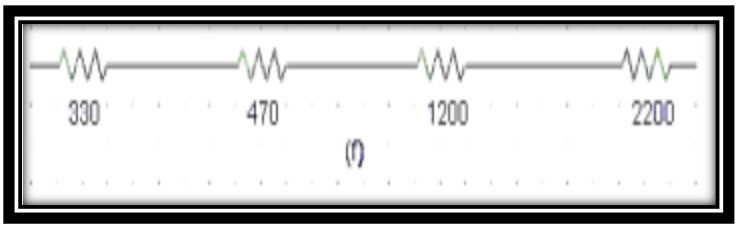
	
Motherboard **R1+R2+R3=330+470+1200=2000**	**Mulimeter: Show equivalence resitence for resistors connected in series**
Rated value: 2k Ω	**Measured Value:1.974k Ω**

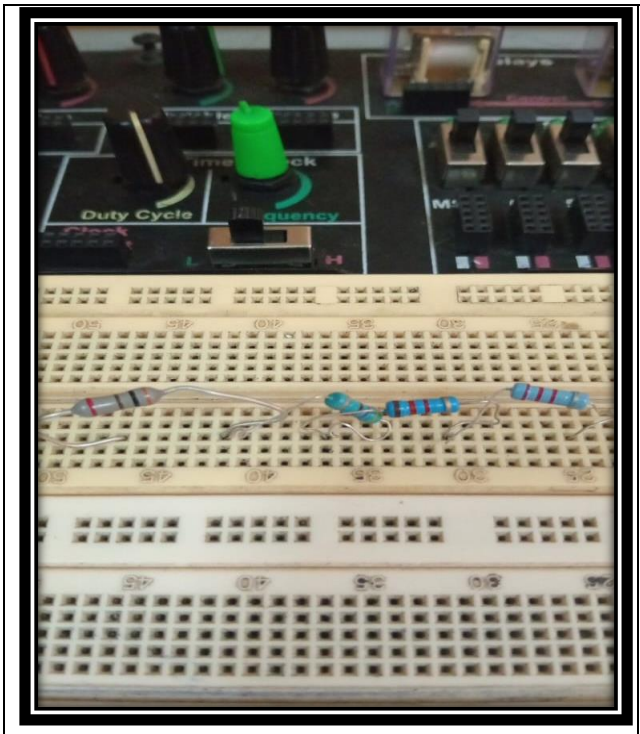
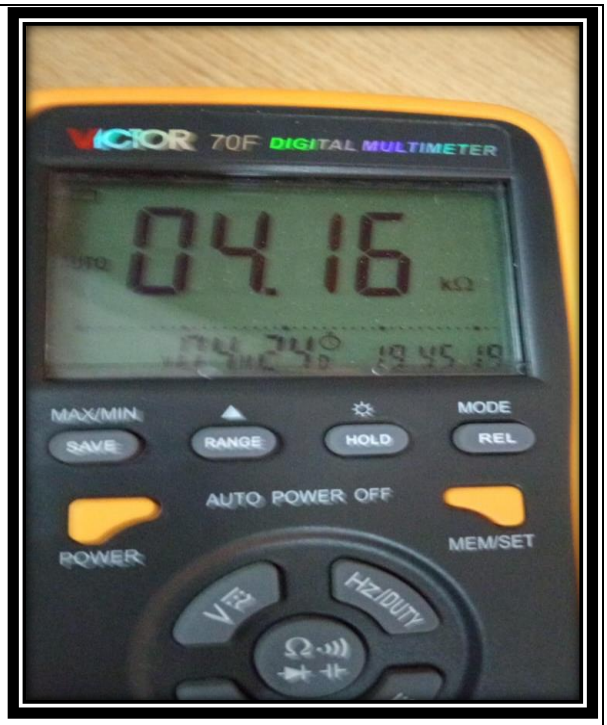
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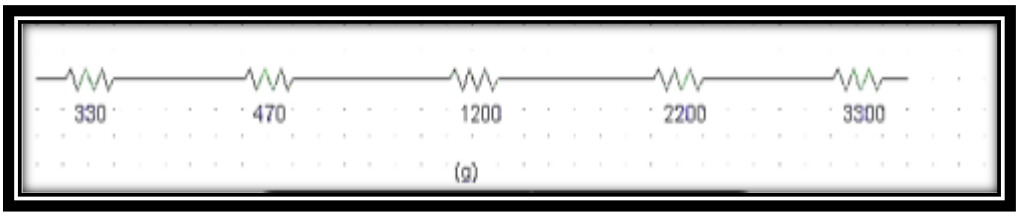
	
Motherboard **R1+R2+R3=1200+2200+3300=6700**	**Mulimeter: Show equivalence resitence for resistors connected in series**
Rated value: 6.7k Ω	**Measured Value:6.70**k Ω

>>>>>>>(f)<<<<<<<<



	
Motherboard **R1+R2+R3+R4=330+470+1200+2200** **=4200**	**Mulimeter: Show equivalence resitence for resistors connected in series**
Rated value: 4.2k Ω	**Measured Value:4.16**k Ω

>>>>>>>(g)<<<<<<<<



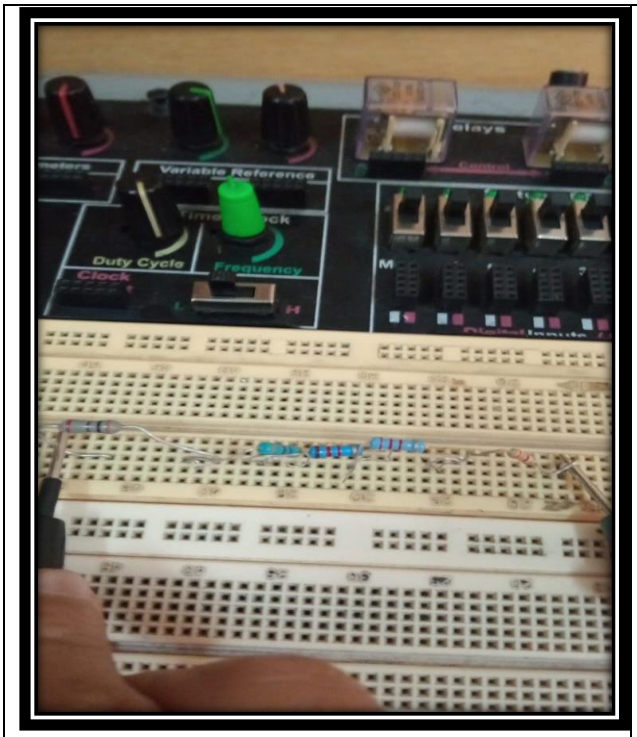
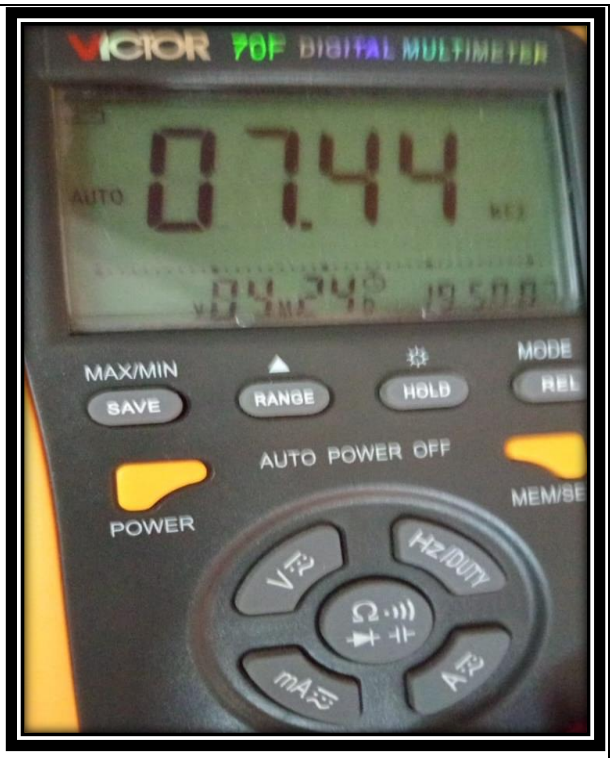
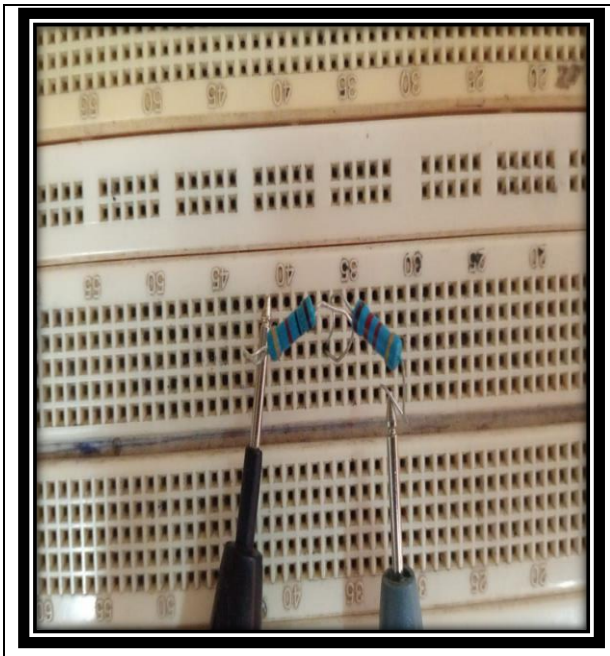
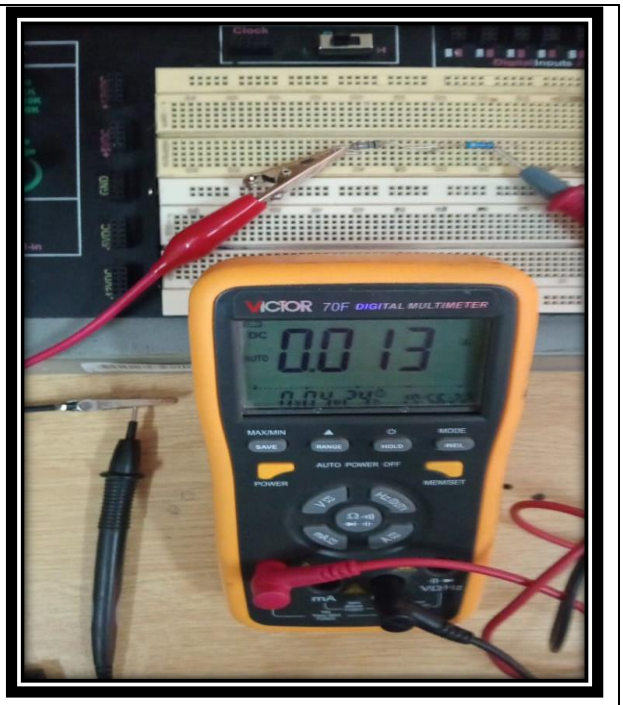
	
Motherboard **R1+R2+R3+R4+R5=** **330+470+1200+2200+3300** **=7600k**	**Mulimeter:Show equivalence resitence for resistors connected in series**
Rated value: 7.6k Ω	**Measured Value:7.44k Ω**

Table of all combinations::

Combinations	R1	R2	R3	R4	R5	R	Measured Value (Ω)	Calculated Value (Ω)	Formula for RT
A	1000	*	1200	*	*	*	2.199KΩ	2.2KΩ	R1+R2...Rn
B	1200	*	2200	*	*	*	3.389KΩ	3.3KΩ	R1+R2...Rn
C	*	*	4700	3300	*	*	8KΩ	7.81KΩ	R1+R2...Rn
D	330	*	470	*	1200	*	2KΩ	1.970KΩ	R1+R2...Rn
E	3300	*	2200	*	1200	*	6.7KΩ	6.70KΩ	R1+R2...Rn
F	330	470	1200	*	2200	*	4.2KΩ	4.16KΩ	R1+R2...Rn
G	330	470	1200	3300	2200	*	7.6KΩ	7.44KΩ	R1+R2...Rn

========(Task 2)==========

	
Motherborad with rsesitors in seriers	**Int this image multimeter is connected ion serieries with resistrors to measure current**



Int this image multimeter is connected ion serieries with resistrors to measure current

Table of all combinations:

Combinations	Applied Voltage (V)	Measured Current (I)	Ohm's Law (R = V/I) (Ω)	Measured Resistance (Ω)	Calculated Resistance (Ω)
A	10V	4.54mA	2.20KΩ	2.199KΩ	2.2KΩ
B	10V	3mA	3.33KΩ	3.389KΩ	3.3KΩ
C	10V	1.3mA	8.0KΩ	8KΩ	7.8KΩ
D	10V	5mA	2.0Ω	2KΩ	1.970KΩ
E	10V	1.49mA	6.71KΩ	6.7KΩ	6.7KΩ
F	10V	2.38mA	4.20KΩ	4.2KΩ	4.16KΩ
G	10V	1.31mA	7.63KΩ	7.6KΩ	7.44KΩ

Conclusion:

In a series circuit, the total resistance (RTR_TRT) is the sum of all individual resistances, causing the overall resistance to increase and the current to decrease according to Ohm's Law (V=IRV = IRV=IR). To measure the current in a series circuit using a digital multimeter (DMM), the circuit must be powered off first. The DMM is then set to the appropriate current (A or mA) setting, and its probes are connected in series by breaking the circuit at any point. Once the circuit is powered on, the DMM displays the current, which remains the same throughout all resistors in the series combination

Connect the Multimeter in Series –

- Plug the **red probe** into the **A or mA** port (depending on the expected current).
- Plug the **black probe** into the **COM** port.
- Place the probes at the two ends of the break in the circuit.

Turn On Power – Switch on the circuit, and the DMM will display the current.

Observe the Reading – In a series circuit, the current remains the **same** throughout all resistor